Supplementary Materials

Enlightenment the Dynamic Behavior of Norbornene–Modified 'Click' 4–arm Polyethylene Glycol Hydrogel: Delving into Framework Properties and Transport Properties through Molecular Dynamics Simulations

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Table S1. System Details

Polymer name	**Initial box dimension (nm^3)**	**Final box dimension (nm^3)**	**Total number of polymer molecules**	**Total number of water molecules**	**Total number of molecules**
PEG	12.26×12.26×12.26	7.08×7.08×7.08	1162	45724	46886
NPEG	12.61×12.61×12.61	6.76×6.76×6.76	1322	39240	40562
TPEG	14.74×14.74×14.74	6.72×6.72×6.72	1466	38316	39782
CPEG	24.51×24.51×24.51	8.83×8.83×8.83	6746	83028	89774

Table S2: The non-bonded parameter for force fields

Atom type	Sigma (nm)	Epsilon (KJ/mol)
CT	3.39967e-01	4.57730e-01
OH	3.06647e-01	8.80314e-01
OS	3.00001e-01	7.11280e-01
H1	2.47135e-01	6.56888e-02
CM	3.39967e-01	3.59824e-01
C	3.39967e-01	3.59824e-01
O	2.95992e-01	8.78640e-01
HA	2.59964e-01	6.27600e-02
HC	2.64953e-01	6.56888e-02
HO	0.00000e+00	0.00000e+00
SH	3.56359e-01	1.04600e+00
HS	1.06908e-01	6.56888e-02

Table S3: The bonded parameter for the force field

Bond name	Bond length (nm)	Bond constant (KJ/mol.nm^2)
CT-CT	1.5260e-01	2.5941e+05
CT-OH	1.4100e-01	2.6778e+05
CT-H1	1.0900e-01	2.8451e+05
OH-HO	9.6000e-02	4.6275e+05
CM-CM	1.5100e-01	2.6527e+05
CM-CT	1.3500e-01	4.5940e+05
C-O	1.2290e-01	4.7698e+0
C-OH	1.3640e-01	3.7656e+05
CT-C	1.5220e-01	2.6527e+05
SH-CT	1.8100e-01	1.9832e+05
SH-HS	1.3360e-01	2.2928e+05

Molecular dynamics equilibration parameters

```
title                = AMBER NPT equilibration

; Run parameters
integrator           = md                        ; leap-frog integrator
tinit                = 0
nsteps               = 62,500,000                ; 0.002 * 62,500,000 = 125000 ps
dt                   = 0.002                     ; 2 fs

; Removing CM Translation and Rotation
comm_mode            = Linear
nstcomm              = 1000
comm-grps            = System
; Output control
nstxout              = 5000
nstvout              = 5000
nstenergy            = 5000
nstlog               = 5000
nstfout              = 5000
nstxtcout            = 5000

; Bond parameters
constraints          = h-bonds
continuation         = yes
constraint_algorithm = lincs
lincs_iter           = 1
lincs_order          = 4
```

; Neighbour Searching

ns_type = grid

nstlist = 20

rlist = 1.0 the short-range neighbor list

; Periodic boundary conditions

pbc = xyz

periodic-molecules = no

; Nonbonded settings

cutoff-scheme = Verlet

rvdw = 1.0

vdwtype = Cut-off

vdw-modifier = Potential-shift-Verlet

rvdw-switch = 0 ;

DispCorr = EnerPres

; Electrostatics

rcoulomb = 1.2

coulombtype = PME

pme_order = 4

fourierspacing = 0.125

; Temperature coupling is on

tcoupl = V-rescale

tc-grps = System

tau_t = 0.1

ref_t = 310

; Pressure coupling is off

pcoupl = Parrinello-Rahman

pcoupltype = Isotropic

nstpcouple = -1

tau-p = 40

compressibility = 4.5e-3

ref-p = 1

; Velocity generation

gen_vel = no

gen_seed = -1

Non-equilibration molecular dynamics parameter

; Run parameters

integrator = md

nsteps = 750000000

dt = 0.002

; Removing CM Translation and Rotation

comm_mode = Linear

nstcomm = 1000

; Output control

nstxout = 5000

nstvout = 5000

nstfout = 5000

nstenergy = 500

nstlog = 5000

nstxout-compressed = 5000

compressed-x-grps = System

; Nonbonded settings

cutoff-scheme = Verlet

ns_type = grid

nstlist = 20

rlist = 1.0

rcoulomb = 1.2

rvdw = 1.0

vdwtype = Cut-off

vdw-modifier = Potential-shift-Verlet

rvdw-switch = 0

DispCorr = EnerPres

; Electrostatics

coulombtype = PME

pme_order = 4

fourierspacing = 0.125

; Temperature coupling is on

tcoupl = V-rescale

tc-grps = System

tau_t = 0.1

ref_t = 310

; Pressure coupling is on

pcoupl = no

pcoupltype = anisotropic

nstpcouple = -1

tau_p = 40.0

ref_p = 1.01325

compressibility = 4.5e-5

; Periodic boundary conditions

pbc = xyz

; Velocity generation

gen_vel = no

;Non equilibrium md

deform = 0 0 0.0002 0 0 0

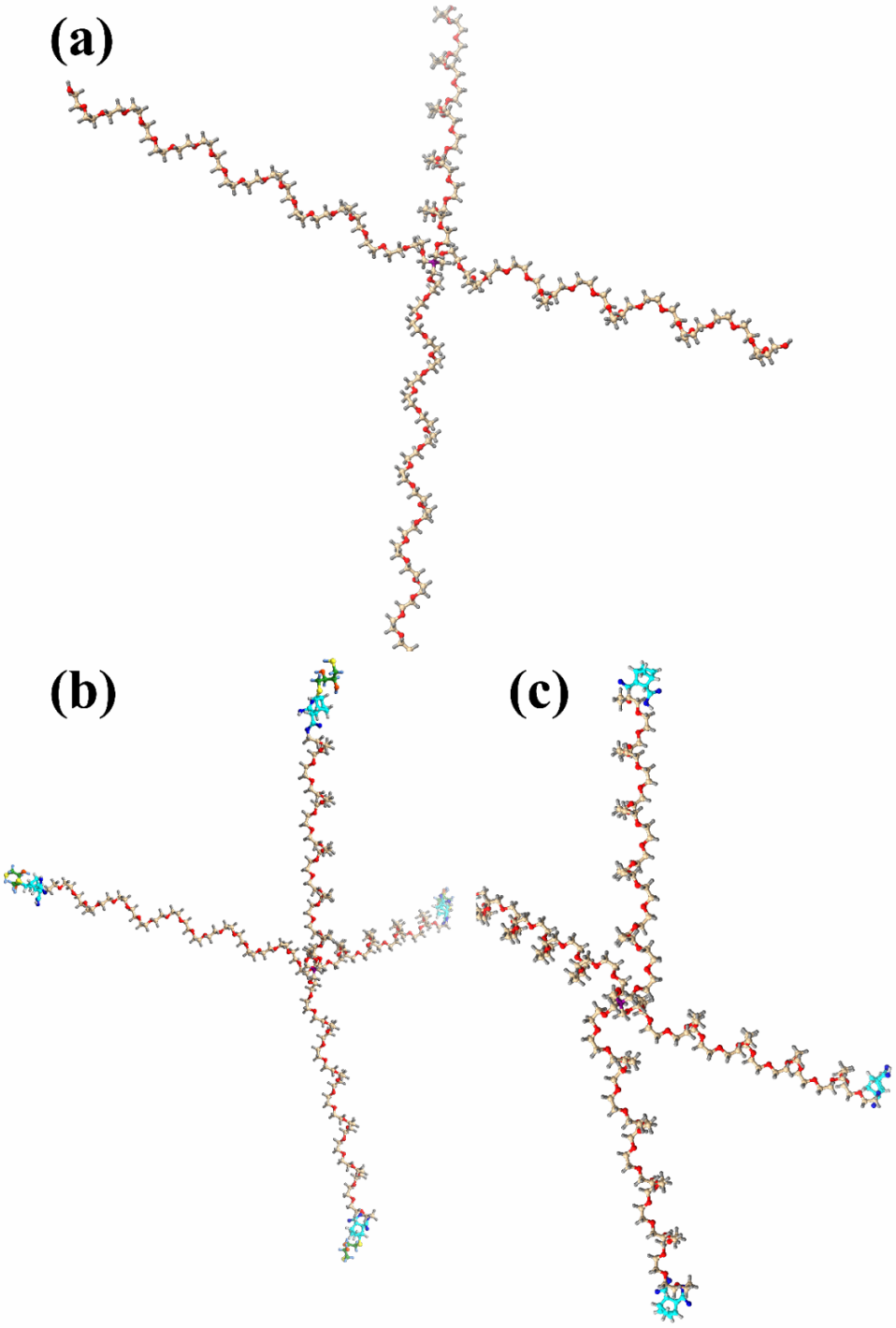


Figure S1. The initial geometry (a) 4-arm PEG, (b) 4-arm NPEG, and (c) 4-arm TPEG polymer chains

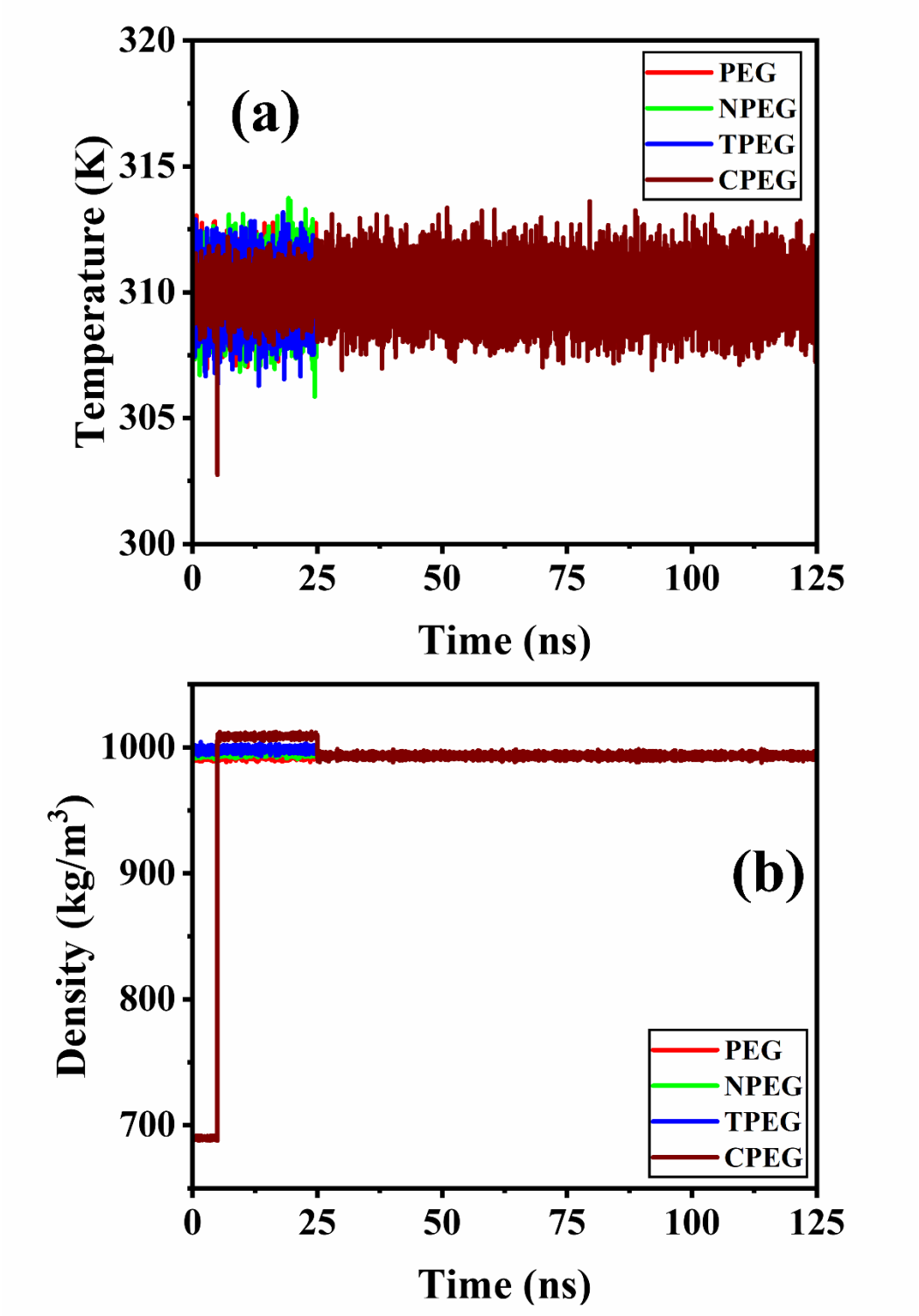


Figure S2. The NPT equilibration data (a) The temperature profile (b) The density profile for all systems

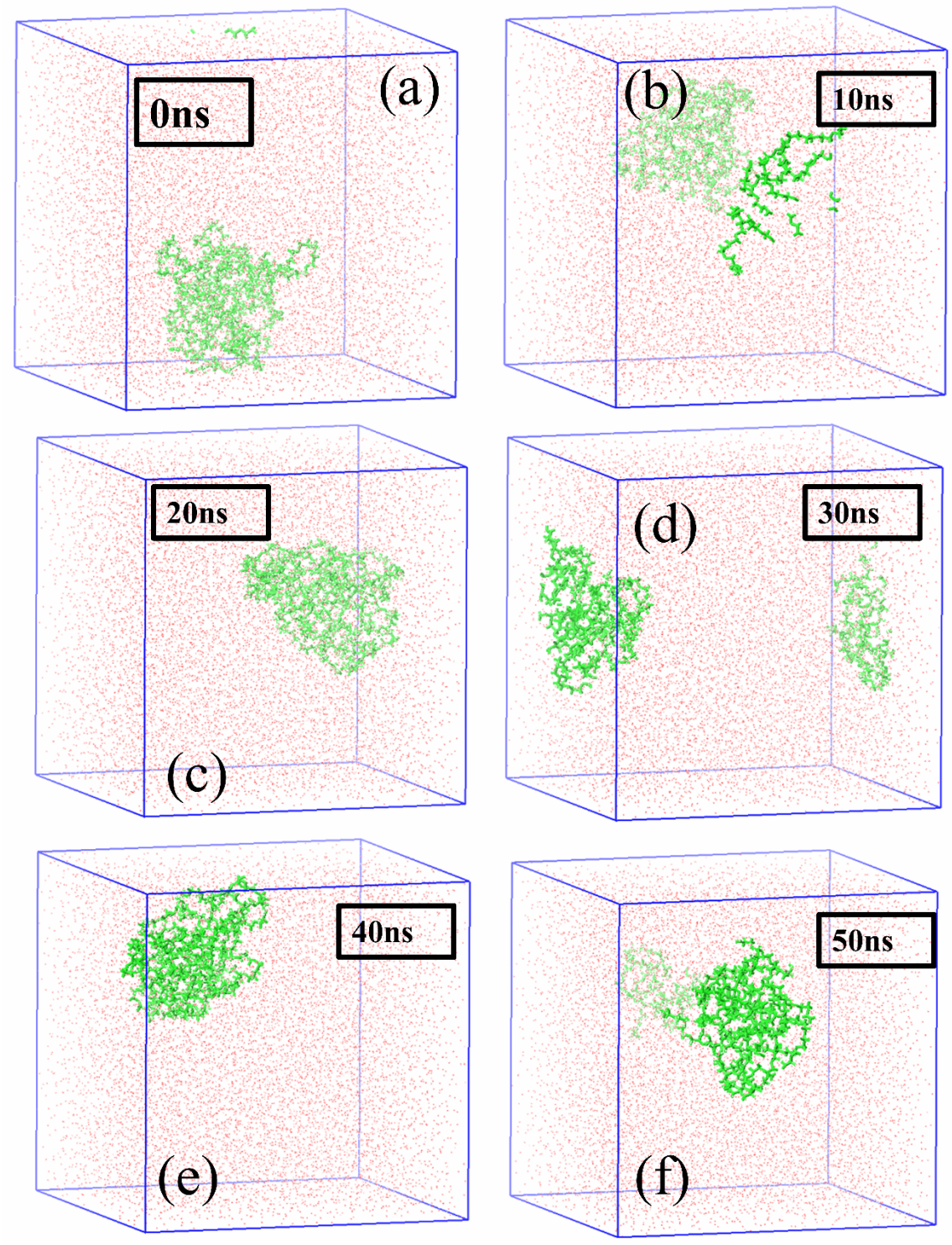


Figure S2. Simulation snapshot of 4-arm PEG hydrogel at the different time intervals (a) 0 ns (b) 10 ns (c) 20 ns (d) 30 ns (e) 40 ns (f) 50 ns

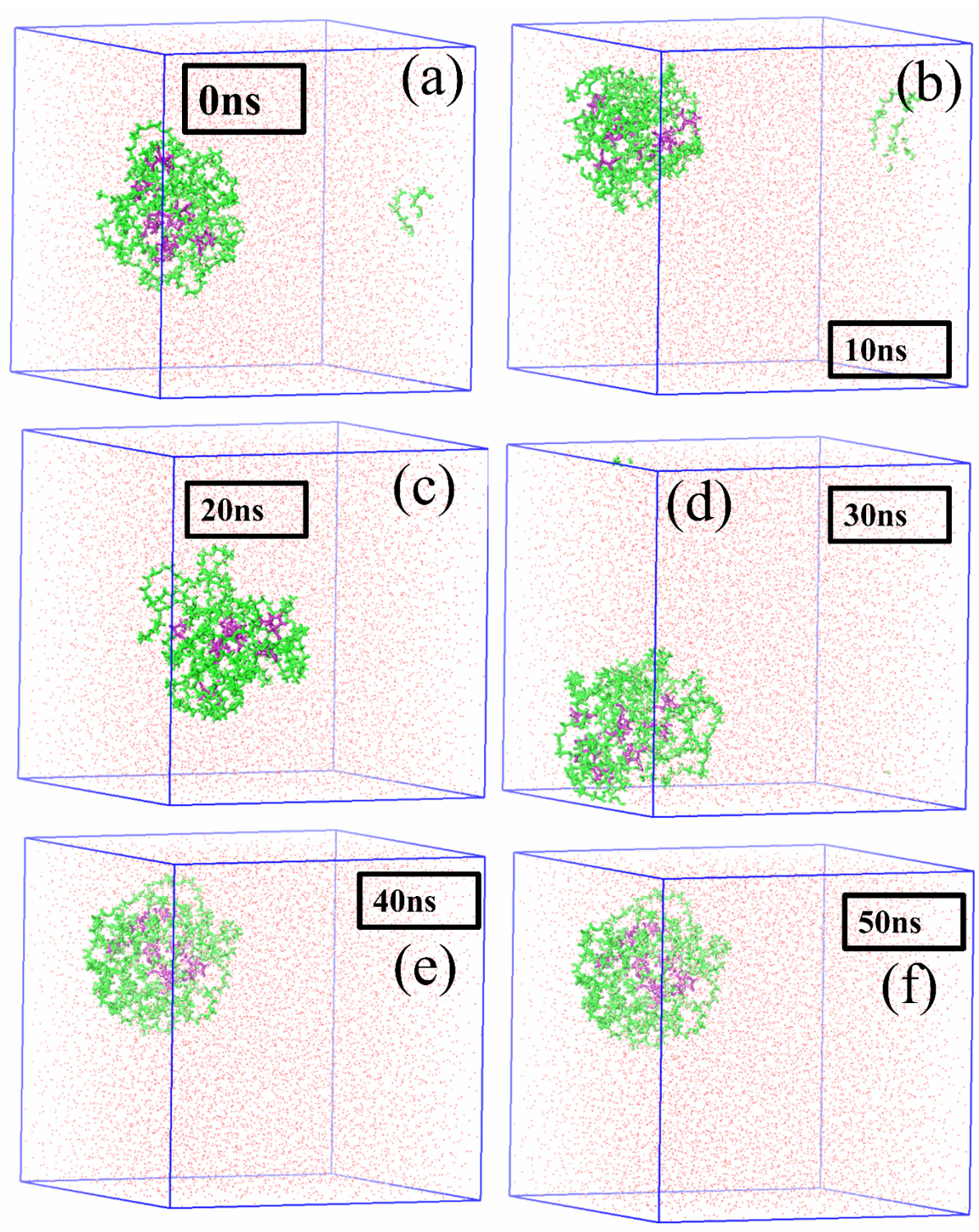


Figure S3. Simulation snapshot of 4-arm NPEG hydrogel at the different time intervals (a) 0 ns (b) 10 ns (c) 20 ns (d) 30 ns (e) 40 ns (f) 50 ns

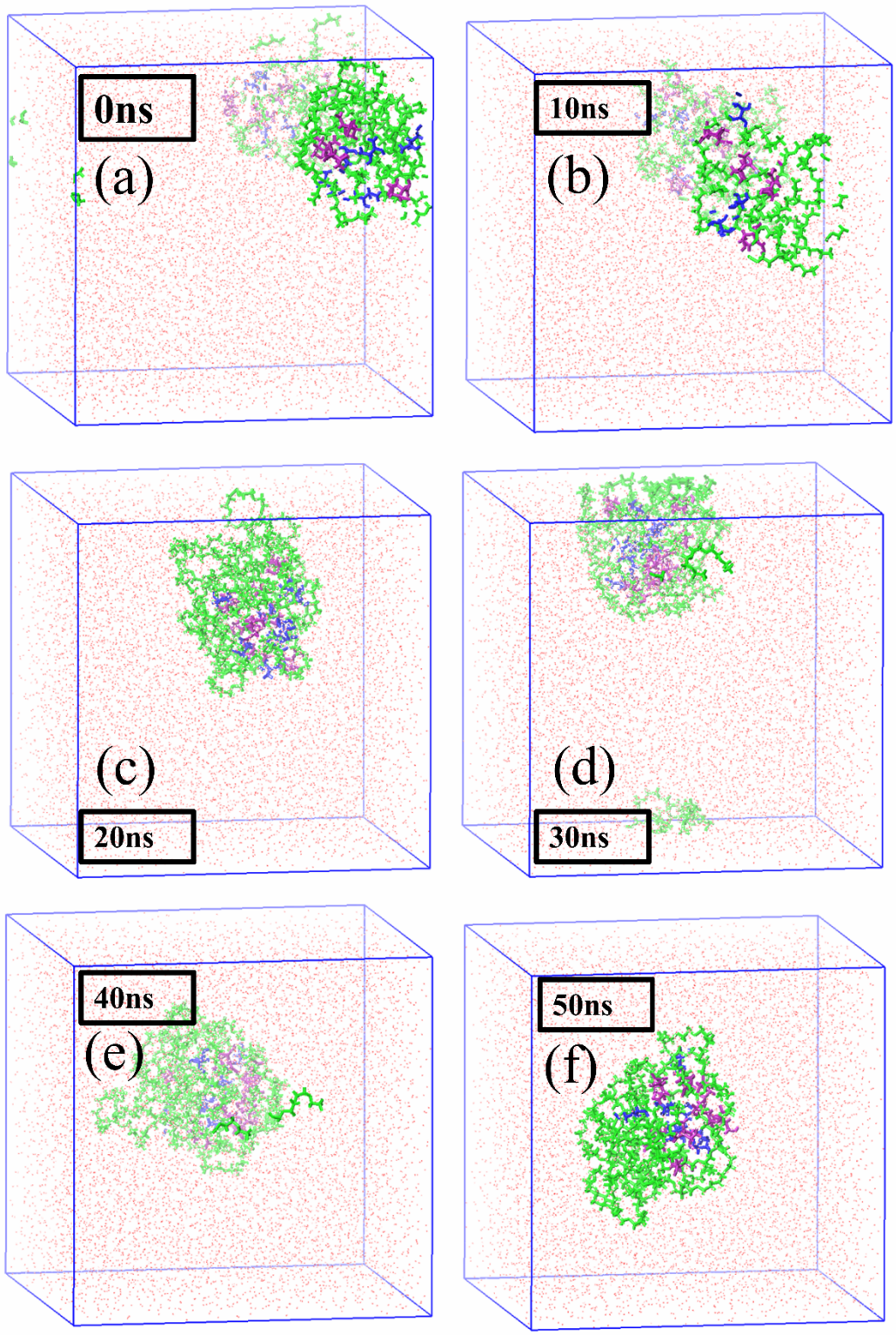


Figure S4. Simulation snapshot of 4-arm TPEG hydrogel at the different time intervals (a) 0 ns (b) 10 ns (c) 20 ns (d) 30 ns (e)40 ns (f) 50 ns

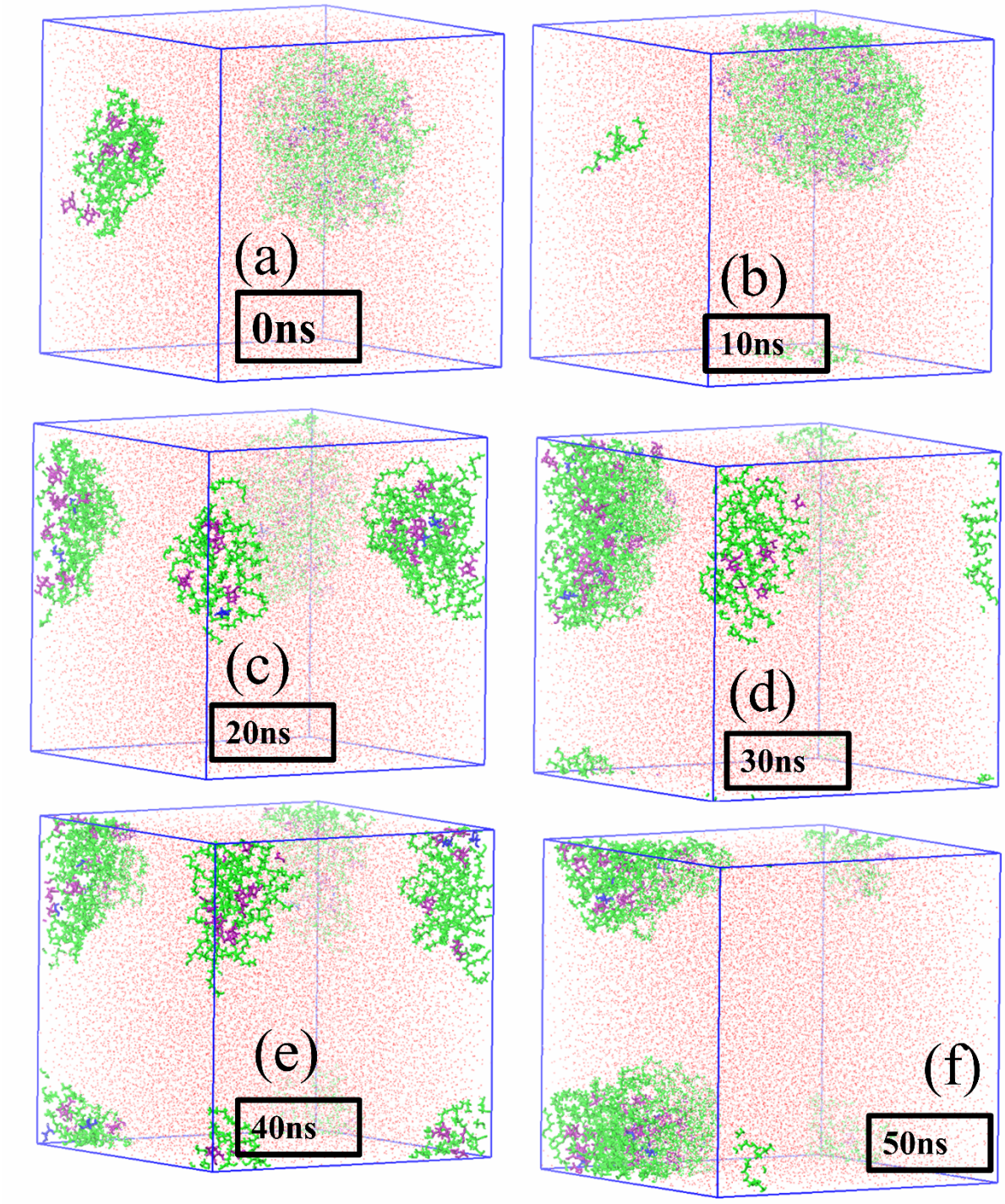


Figure S5. Simulation snapshot of 4-arm CPEG hydrogel at the different time intervals (a) 0 ns (b) 4 ns (c) 8 ns (d) 12 ns (e) 16 ns (f) 20 ns